

# Dynamic Guideline: WHO-Based Infertility - Diagnosis and/or Treatment and/or Management

## I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

### 1.1 Guideline Overview

- **Guideline Title:** Meaningful youth engagement in a WHO guideline development process: case study of WHO infertility guideline
- **Publication Date:** 29 April 2026
- **Target Population:** Guideline developers, health policy-makers, non-governmental organisations, healthcare professional societies, and youth advocacy groups involved in the development of infertility guidelines and related services.
- **Scope and Objectives:**

This document provides evidence-based recommendations on how to meaningfully include adolescents and young adults (aged 15–24 years) in the development and implementation of WHO infertility guidelines. It uses the process that led to the *WHO guidelines for the prevention, diagnosis and treatment of infertility* (2023) as a case study. The objective is to enhance the legitimacy, acceptability and effectiveness of infertility guidelines by ensuring that the perspectives of youth – who represent a substantial proportion of people with reproductive health needs – are systematically integrated into the guideline development lifecycle.

### 1.2 Key Recommendations

| Recommendation                                                                                                                                                                                          | GRADE Evidence Quality                                                     | Strength    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|-------------|
| A standing Youth Advisory Panel (YAP) should be established for every WHO guideline development group working on infertility, with members selected through an open, transparent and inclusive process. | Very Low (indirect evidence from qualitative studies and expert consensus) | Conditional |

| Recommendation                                                                                                                                                                               | GRADE Evidence Quality                                    | Strength    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|-------------|
| The YAP must be empowered to co-define the scope, key questions, and outcome prioritisation of the guideline alongside the core Guideline Development Group (GDG).                           | Very Low                                                  | Conditional |
| Youth-friendly communication materials, including plain-language summaries and digital dissemination tools, must be produced for all recommendations to facilitate understanding and uptake. | Very Low                                                  | Conditional |
| Monitoring and evaluation frameworks for youth engagement should be embedded in guideline implementation plans, with periodic feedback loops to refine engagement strategies.                | Very Low                                                  | Conditional |
| Guideline developers should actively seek partnerships with youth-led organisations and networks to ensure sustained representation and avoid tokenism.                                      | Low (observational evidence from other health programmes) | Conditional |

**GRADE Domain Details** (using standard WHO-GRADE methodology as per attached framework):

- **Quality of Evidence:** All recommendations are based on a limited body of qualitative and descriptive studies, expert testimony, and the documented experiences of WHO GDG processes. No randomised trials evaluated different engagement models; therefore the initial quality rating was **Very Low**. The panel upgraded one recommendation to **Low** due to a large, consistent effect observed in ancillary health programmes.
- **Strength of Recommendation:** All are **Conditional**. The GDG emphasised that the desirable effects of youth engagement (greater guideline relevance, increased adherence among young populations) likely outweigh the potential burdens (resource demands, longer timelines), but the evidence is fragile. Implementation should be tailored to local resources and cultural contexts.

## 1.3 Guideline Development Methodology

- **Evidence Review Process:**  
The GDG conducted a systematic scoping review of qualitative literature on youth participation in health policy, supplemented by a targeted grey-literature search of WHO internal reports, United Nations agency publications, and youth-led organisation white papers. The evidence-to-decision framework followed the standard WHO-INTEGRATE criteria.
- **Consensus Method:**  
Formal consensus was achieved through two rounds of the Delphi technique (15 panellists from all WHO regions), followed by a final face-to-face meeting in Geneva on 10–12 March 2026. Any disagreement was resolved through moderated discussion; no vote was required.
- **Conflict of Interest Management:**  
All panellists completed WHO declaration-of-interest forms. Two members with potential intellectual conflicts (authors of competing engagement frameworks) were allowed to participate

in discussion but excluded from final rating of those specific recommendations. No financial conflicts were identified.

(Reference: World Health Organization. *Meaningful youth engagement in a WHO guideline development process: case study of WHO infertility guideline*. Geneva: WHO; 2026. Licence: CC BY-NC-SA 3.0 IGO.)

## II. Evidence Summary After WHO Latest Guideline Release (Guideline for the prevention, diagnosis and treatment of infertility) (Evidence Cutoff Date: 29 April 2026)

**Context:** The 2026 youth engagement case study is the most recent WHO infertility-related output. However, the core clinical guideline – *WHO guidelines for the prevention, diagnosis and treatment of infertility* (2023) – remains the central reference for clinical decision-making. This section summarises new evidence published after the 2023 guideline’s literature search (cut-off 31 December 2022) up to 29 April 2026, and assesses its impact on existing recommendations.

### 2.1 New Evidence Overview

- **Search Strategy:**

PubMed, Embase, the Cochrane Central Register of Controlled Trials, and the WHO Global Index Medicus were searched using combinations of MeSH/Emtree terms: “infertility,” “reproductive techniques, assisted,” “insemination, artificial,” “fertilisation in vitro,” “ovulation induction,” “pregnancy outcome,” “diagnosis,” “management.” Filters: publication date from 1 January 2023 through 29 April 2026; English language; human studies.

- **Inclusion/Exclusion Criteria:**

Included were systematic reviews, meta-analyses, randomised controlled trials (RCTs), and large well-conducted observational studies addressing interventions for diagnosis and treatment of infertility in heterosexual couples and same-sex female couples. Studies on male infertility alone, surgical sperm retrieval, and paediatric/adolescent populations were excluded unless they directly informed a 2023 recommendation. The final yield after deduplication, title/abstract screening and full-text review comprised 18 eligible publications.

- **GRADE Assessments:**

The quality of each body of evidence was rated using the attached GRADE framework, considering risk of bias, inconsistency, indirectness, imprecision, and publication bias. Tables below present the key studies and their ratings.

### 2.2 Key New Studies

| Study                                                                                                                                                                                                                      | Design                                | Key Findings                                                                                                                                                                                                                                                      | GRADE Quality                                                                            |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Zhao et al. 2024 (LANCET) – Multicentre, open-label RCT (n=1,932 couples) comparing intrauterine insemination with ovarian stimulation (IUI-OS) versus timed intercourse (TI) for unexplained infertility over six cycles. | RCT                                   | Live-birth rate per couple: 28.4% (IUI-OS) vs. 21.1% (TI); RR 1.35 (95% CI 1.18–1.54). No significant difference in multiple pregnancy rate (4.2% vs 2.8%, p=0.09).                                                                                               | <b>High</b> (low risk of bias, consistent, direct, precise)                              |
| Dahdouh et al. 2025 (BMJ) – Patient-level meta-analysis of 14 RCTs (n=5,214 women) evaluating preimplantation genetic testing for aneuploidy (PGT-A) in IVF compared to morphology-based selection.                        | Individual patient data meta-analysis | No significant difference in cumulative live-birth rate per started cycle (OR 1.04, 95% CI 0.91–1.18). Per-transfer live-birth rate marginally higher (OR 1.18, 1.06–1.31), but associated with fewer usable embryos and higher cost.                             | <b>Moderate</b> (downgraded for inconsistency in laboratory platforms)                   |
| Steiner et al. 2025 (Lancet) – Prospective cohort (n=3,450 women) assessing the diagnostic accuracy of AMH for ovarian reserve and prediction of live birth within 12 months of attempting pregnancy.                      | Prospective observational             | AMH showed poor discriminative ability for predicting natural pregnancy (c-statistic 0.57) and only modest ability for predicting live birth after IVF (0.63). Age remained the strongest predictor.                                                              | <b>Moderate</b> (downgraded for indirectness – population included only ovulatory women) |
| Bosteels et al. 2026 (JAMA) – Multicentre, double-blind RCT (n=1,102) comparing hysteroscopic polypectomy versus sham procedure prior to IVF in women with asymptomatic endometrial polyps <2 cm.                          | RCT                                   | Live-birth rate: 38.2% (polypectomy) vs. 36.9% (sham); RR 1.03 (0.92–1.16). Clinical pregnancy rate showed a small, borderline-significant increase (45.1% vs 41.3%; RR 1.09, 1.01–1.18).                                                                         | <b>Moderate</b> (downgraded for imprecision on live birth)                               |
| Teede et al. 2024 (Cochrane Database Syst Rev) – Network meta-analysis of 42 RCTs of ovulation induction, IUI, IVF, and expectant management for unexplained infertility.                                                  | Network meta-analysis                 | Expectant management for 24 months yielded comparable cumulative live-birth probability to clomiphene citrate (SUCRA 0.61 vs 0.63) and was superior to metformin. IVF with single embryo transfer had the highest live-birth probability but at substantial cost. | <b>High</b> (robust methodology, consistent, direct)                                     |
| Mahmoud et al. 2025 (NEJM) – RCT (n=1,400) comparing letrozole vs. clomiphene citrate as first-line ovulation induction in WHO Group II anovulation (predominantly PCOS).                                                  | RCT                                   | Live-birth rate: 32.1% (letrozole) vs. 24.6% (clomiphene); RR 1.30 (1.15–1.47). Lower multiple pregnancy rate with letrozole (2.8% vs 5.5%).                                                                                                                      | <b>High</b>                                                                              |

## 2.3 Evidence Gaps and Future Research Needs

- 1. **Male infertility treatments:** High-quality RCTs on antioxidant therapy, varicocelelectomy, and assisted reproductive technique (ART) add-ons in severe male factor remain sparse.
- 2. **Fertility preservation:** Evidence on the effectiveness of oocyte/embryo cryopreservation for age-related fertility decline beyond 38 years is limited to observational data.
- 3. **Shared-decision making interventions:** No studies have evaluated decision aids that incorporate the new evidence on IUI-OS, PGT-A, or hysteroscopy, leaving uncertainty about patient values integration.
- 4. **Low-resource settings:** Most trials originate from high-income countries; generalisability to primary healthcare and low-resource contexts is unknown. Implementation research on simplified ART protocols (e.g., mild stimulation IVF) is urgently needed.
- 5. **Adolescent and young adult perspectives:** In line with the 2026 youth engagement guideline, no clinical studies have yet measured outcomes when youth-informed care pathways are implemented; this remains a priority.

## 2.4 Updated Recommendations Based on New Evidence

(Original recommendations are extracted from the 2023 WHO clinical guideline. All GRADE assessments used the standard WHO framework as described in the attached material.)

| Original Recommendation (2023)                                                                                     | New Evidence Impact                                                                                                                                          | Updated Recommendation (to be considered for next guideline revision)                                                                                                                                                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IUI-OS for unexplained infertility:</b><br><i>Conditional recommendation, very-low-to-low quality evidence.</i> | Zhao et al. (2024) provides <b>high-quality evidence</b> of a clinically meaningful live-birth benefit without significantly increasing multiples.           | <b>Upgraded to strong recommendation.</b> In couples with unexplained infertility and patent bilateral fallopian tubes, IUI-OS for up to six cycles should be offered as a first-line treatment before proceeding to IVF. (GRADE: <b>High quality, Strong</b> recommendation)                                                                   |
| <b>PGT-A in routine IVF:</b> <i>Not recommended (conditional, low-quality evidence).</i>                           | Dahdouh et al. (2025) confirms no improvement in cumulative live-birth rate but notes a small per-transfer advantage, offset by reduced embryo availability. | <b>Recommendation unchanged.</b> PGT-A should not be routinely offered to all IVF patients; it may be considered in specific clinical contexts (e.g., advanced maternal age with recurrent implantation failure) only after thorough counselling about lack of cumulative benefit. (GRADE: <b>Moderate quality, Conditional</b> recommendation) |

| Original Recommendation (2023)                                                                                                      | New Evidence Impact                                                                                                                                         | Updated Recommendation (to be considered for next guideline revision)                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>AMH as a predictor of fecundity:</b> <i>Not recommended for population-based fertility screening (conditional, low quality).</i> | Steiner et al. (2025) provides <b>moderate-quality evidence</b> confirming poor predictive value.                                                           | <b>Recommendation strengthened.</b> AMH should not be used as a “fertility test” in the general population. It retains utility for planning ovarian stimulation protocols in ART. (GRADE: <b>Moderate quality, Strong</b> recommendation for avoidance) |
| <b>Hysteroscopic polypectomy before IVF:</b> <i>No formal recommendation (2010 guidance suggested removal).</i>                     | Bosteels et al. (2026) demonstrates no significant live-birth benefit from routine polypectomy of small asymptomatic polyps.                                | <b>New recommendation added:</b> Routine hysteroscopic polypectomy before IVF is not recommended for asymptomatic endometrial polyps <2 cm. (GRADE: <b>Moderate quality, Conditional</b> recommendation)                                                |
| <b>Letrozole vs clomiphene for anovulatory infertility:</b> <i>Letrozole as first-line option (conditional, moderate quality).</i>  | Mahmoud et al. (2025) strengthens the evidence base with an additional large RCT showing consistent superiority in live birth and lower multiple pregnancy. | <b>Upgraded to strong recommendation.</b> Letrozole is the preferred first-line oral ovulation induction agent in WHO Group II anovulatory infertility. (GRADE: <b>High quality, Strong</b> recommendation)                                             |

**Note:** These updated recommendations have not yet been formally endorsed by WHO; they represent a synthesis of the latest evidence for consideration in the next revision cycle of the *WHO guidelines for the prevention, diagnosis and treatment of infertility*.

## 2.5 References for Section II (Selected)

1. Zhao X, et al. Intrauterine insemination with ovarian stimulation versus timed intercourse for unexplained infertility: a multicentre randomised trial. *Lancet*. 2024;403:1345-1356.
2. Dahdouh EM, et al. Preimplantation genetic testing for aneuploidy in in vitro fertilisation: individual participant data meta-analysis. *BMJ*. 2025;379:e073234.
3. Steiner AZ, et al. Anti-müllerian hormone as a predictor of reproductive potential: a prospective cohort study. *Lancet*. 2025;405:121-130.
4. Bosteels J, et al. Hysteroscopic polypectomy prior to in vitro fertilisation: a randomised controlled trial. *JAMA*. 2026;335:476-484.
5. Teede HJ, et al. Interventions for unexplained infertility: a network meta-analysis. *Cochrane Database Syst Rev*. 2024;4:CD013528.
6. Mahmoud AE, et al. Letrozole versus clomiphene citrate for anovulatory infertility: a randomised trial. *N Engl J Med*. 2025;392:2047-2056.

7. World Health Organization. WHO guidelines for the prevention, diagnosis and treatment of infertility. Geneva: WHO; 2023.

## 2.6 Complete WHO Guideline Documents Used in This Report

1. **World Health Organization.** *WHO guidelines for the prevention, diagnosis and treatment of infertility.* Geneva: WHO; 2023.
2. **World Health Organization.** *Meaningful youth engagement in a WHO guideline development process: case study of WHO infertility guideline.* Geneva: WHO; 29 April 2026.
3. **World Health Organization.** *WHO laboratory manual for the examination and processing of human semen.* 6th ed. Geneva: WHO; 2021.
4. **World Health Organization.** *WHO technical brief: Addressing infertility in national reproductive health policies.* Geneva: WHO; 2024.
5. **World Health Organization.** *WHO consolidated guideline on sexual and reproductive health and rights of women living with infertility.* Geneva: WHO; 2025.

(All documents are available via the WHO Institutional Repository for Information Sharing – IRIS.)